MSG No. 137.509 $P4_2/n'mc$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,4,5,6,11,12,14,15]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,7,8,9,10,13,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,4,5,6,11,12,14,15]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,3,7,8,9,10,13,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[9,10,13,16]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[11,12,14,15]`

Table 4: Wyckoff site: `4d`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[2,3,7,8]`
3	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[9,10,13,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[11,12,14,15]`

Table 5: Wyckoff site: `8e`, site symmetry: `-1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,13]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,14]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,15]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,16]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[5,9]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[6,10]`
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[7,11]`
8	$\left[0, 0, \frac{1}{2}\right]$	`[8,12]`

Table 6: Wyckoff site: `8f`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[1,12]`
2	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	`[2,9]`
3	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[3,10]`
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[4,11]`
5	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	`[5,14]`
6	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[6,15]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[7,16]`
8	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[8,13]`

Table 7: Wyckoff site: `8g`, site symmetry: `.m.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,5]`
2	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[2,7]`
3	$\left[y,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[3,8]`
4	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	`[4,6]`
5	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[9,13]`
6	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	`[10,16]`
7	$\left[y+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[11,14]`
8	$\left[-y,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[12,15]`

Table 8: Wyckoff site: `16h`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\ y,\ z\right]$	`[5]`
6	$\left[x,\ \frac{1}{2}-y,\ z\right]$	`[6]`
7	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	`[8]`
9	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[9]`
10	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	`[10]`
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11]`
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[12]`
13	$[-x,\ -y,\ -z]$	`[13]`
14	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[14]`
15	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[15]`
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[16]`